# Hierarchical Shell Manifold Theory

Graded Geometry, Intergrade Dynamics, and Effective Physics

Vincent Mark Garrett
Independent Researcher, Lubbock, Texas, USA
vince1975.garrett@outlook.com

October 2026

**Abstract**

Hierarchical Shell Manifold Theory (HSMT) is a framework for describing a physical system as a hierarchy of geometric sectors that can interact with one another. The word *shell* does not mean an ordinary spherical layer in space. It means a distinct level, or grade, in an underlying geometric organization. Each grade can carry its own state space and geometric data, while the full theory allows information and dynamics to pass between grades.

The central mathematical idea is straightforward. The full state space is separated into grade-labelled pieces. A self-adjoint operator governs the complete dynamics. If attention is restricted to one grade, the other grades can be removed from the description by exact projection. Their influence does not simply disappear: it returns as energy-dependent corrections, memory terms, and effective interactions in the selected sector. In this sense, physics that looks complicated inside one grade may partly reflect coupling to structure that is not directly observed there.

This paper defines the graded arena, explains what a shell is, describes intergrade coupling and exact projection, introduces scale-dependent dimensional observables, and explains how locality, metric structure, time, internal degrees of freedom, gravity-like response, and optional cycle modes can be incorporated. The paper also states the present physical boundary clearly. HSMT does not yet identify which mathematical grading is the grading realized by Nature, and it does not yet provide a uniquely HSMT-specific measurable law connecting physically identified grades. HSMT is therefore a mathematically controlled framework for graded geometry and effective physics, with several conditional physical modules, rather than a completed fundamental theory.

**The theory in one sentence**

HSMT asks whether some of the physics seen inside a familiar sector can be understood as the effective result of a larger geometric hierarchy whose sectors are dynamically coupled.

## Contents

**1 A reader's map of HSMT** **4**

**2 The graded geometric arena** **4**

2.1 What is a manifold? . . . . . 4

2.2 What is a shell? . . . . . 5

2.3 No preferred zero of the grade label . . . . . 5

**3 The state space of the hierarchy** **5**

**4 The microscopic dynamics** **6**

4.1 One generator for the full system . . . . . 6

4.2 Why a bounded hopping term is mathematically convenient . . . . . 7

**5 A useful reference model for intergrade coupling** **7**

**6 What a single grade sees: exact projection** **7**

**7 Memory and delayed influence** **8**

**8 Dimension in HSMT** **9**

8.1 Grade is not dimension . . . . . 9

8.2 Operational dimension from scaling . . . . . 9

**9 Locality: what counts as nearby?** **9**

**10 Metric, measure, and energy** **10**

**11 Time and temporal evolution** **10**

**12 Refinement and changing descriptions** **11**

**13 Internal algebra and matter-like structure** **11**

**14 A controlled gravity-response branch** **11**

**15 The optional P-CYCLE sector** **12**

**16 How HSMT could connect to experiment** **13**

**17 What would make the grading physical?** **13**

**18 What would count as a distinct HSMT prediction? 14**

**19 What HSMT presently says about familiar physics 14**

19.1 Four-dimensional spacetime . . . . . 14

19.2 The Standard Model . . . . . 14

19.3 Gravity . . . . . 15

19.4 Cosmology . . . . . 15

**20 What HSMT does provide now 15**

**21 A worked conceptual example 16**

**22 How to read future HSMT models 16**

**23 Current physical boundary 17**

**24 Conclusion 17**

**A Plain-language glossary 17**

**B Compact mathematical summary 18**

## 1. A reader's map of HSMT

The easiest way to understand HSMT is to separate three questions.

**First: what exists mathematically?** There is a collection of grades. Each grade carries a state space, and the full state space is their direct sum.

**Second: what happens dynamically?** A single microscopic generator acts on the full state space. It may contain terms that keep a state within one grade and terms that move amplitude between grades.

**Third: what does a restricted observer see?** If only one grade, or one group of grades, is accessible to measurement, exact projection produces an effective description for that restricted sector. Hidden grades can then appear indirectly through corrections, resonances, memory, or scale-dependent behavior.

Schematically,

$$\boxed{\text{graded geometry} \longrightarrow \text{intergrade dynamics} \longrightarrow \text{projected effective physics}} \tag{1}$$

That three-step structure is the central architecture of HSMT.

**What HSMT is not**

HSMT is not the statement that the universe is literally made of nested spherical shells. It is not the statement that a grade number is automatically a spatial dimension. It is not a claim that every hidden sector is a new universe. "Shell" is a name for a regime in a hierarchy, and its physical meaning has to be supplied by the full theory and, ultimately, by experiment.

## 2. The graded geometric arena

### 2.1. What is a manifold?

In ordinary geometry, a *manifold* is a space that looks locally like a familiar coordinate space. A line is one-dimensional, a surface is locally two-dimensional, and ordinary spacetime is commonly modeled by a four-dimensional manifold. HSMT uses the word manifold in this general geometric sense, but it allows more than one geometric sector to participate in the full construction.

The public geometric picture is written as

$$\mathcal{M} = \bigsqcup_N \mathcal{M}_N. \tag{2}$$

Here $\mathcal{M}_N$ denotes the geometric sector associated with grade $N$, and the symbol $\bigsqcup$ means that the sectors are treated as distinct pieces of a larger graded arena.

The grade label is a bookkeeping coordinate for the hierarchy. It tells us which sector we are referring to. It does *not* by itself tell us the physical dimension of that sector.

$$\boxed{N \neq d} \tag{3}$$

This distinction is basic. A label such as $N = 4$ is a grade label unless an independent physical rule turns it into something more.

### 2.2. What is a shell?

A useful plain-language definition is:

---

**Shell**

An HSMT shell is a coherent regime in the underlying hierarchy. It is called a shell because it occupies a distinct level of organization, not because it must be a round surface surrounding another surface in ordinary space.

---

A shell may be represented by a manifold $\mathcal{M}_N$, by a Hilbert-space sector $\mathcal{H}_N$, or by both, depending on which part of the theory is being discussed.

The word *grade* emphasizes the ordering of the hierarchy. The word *shell* emphasizes the interpretation of one grade as a coherent physical or geometric regime. They are related ideas, but the grade is the cleaner mathematical label.

### 2.3. No preferred zero of the grade label

The mathematical hierarchy need not come with an absolute origin. If integer labels are used, a relabelling

$$N \mapsto N + c \tag{4}$$

can leave the abstract structure unchanged. This means that absolute grade numbers do not become physical merely because they are written as integers.

What can matter are relations such as adjacency, separation, ordering, or coupling strength between grades.

## 3. The state space of the hierarchy

Quantum states are described in a Hilbert space. A Hilbert space is a vector space equipped with an inner product, which is the mathematical structure used to calculate amplitudes and probabilities.

HSMT separates the total Hilbert space into grade-labelled sectors:

$$\mathcal{H} = \bigoplus_{\gamma \in \Gamma} \mathcal{H}_\gamma. \tag{5}$$

The symbol $\oplus$ means a direct sum. A state in the full theory may therefore have components in several grades at once.

For each grade there is an orthogonal projector $P_\gamma$. A projector is an operator that extracts the part of a state belonging to a chosen sector. The projectors satisfy

$$P_\gamma P_{\gamma'} = \delta_{\gamma\gamma'} P_\gamma, \qquad \sum_\gamma P_\gamma = I. \tag{6}$$

In plain language: two different grade projectors do not overlap, and taken together they recover the whole state space.

A general state can be written as

$$|\Psi\rangle = \sum_\gamma |\Psi_\gamma\rangle, \qquad |\Psi_\gamma\rangle = P_\gamma |\Psi\rangle. \tag{7}$$

Nothing in this decomposition requires a state to remain permanently in one grade. That depends on the dynamics.

## 4. The microscopic dynamics

### 4.1. One generator for the full system

The full dynamics are generated by a self-adjoint operator

$$H = H^\dagger. \tag{8}$$

Self-adjointness is the mathematical condition that allows the usual unitary quantum time evolution

$$U(t) = e^{-itH}. \tag{9}$$

Unitary evolution preserves total probability.

Relative to the grade decomposition, the operator can be written schematically as

$$H = \sum_\gamma P_\gamma H P_\gamma + \sum_{\gamma \neq \gamma'} P_\gamma H P_{\gamma'}. \tag{10}$$

The first sum contains *intra-grade* terms: dynamics that stay within one grade. The second contains *intergrade* terms: dynamics that couple different grades.

**Intergrade coupling**

Intergrade coupling means that the full equations allow a state or influence associated with one grade to affect another grade. If all intergrade terms vanished, the grades would evolve independently and the hierarchy would be dynamically disconnected.

### 4.2. Why a bounded hopping term is mathematically convenient

One useful reference realization separates the operator into a well-controlled diagonal part and a bounded coupling part,

$$H = H_0 + V. \tag{11}$$

If $H_0$ is self-adjoint and $V$ is bounded and symmetric, standard perturbation results preserve self-adjointness of the sum under the usual hypotheses [3]. This gives a clean mathematical way to introduce finite-strength hopping between grades without losing control of the quantum dynamics.

The important physical point is simpler than the theorem: coupling grades does not require abandoning ordinary probability-conserving quantum evolution.

## 5. A useful reference model for intergrade coupling

HSMT does not require one unique coupling law at the framework level. A particularly simple model uses a Gaussian profile,

$$K_{NM} = K_0 \exp\left[-\frac{(N-M)^2}{2\sigma^2}\right]. \tag{12}$$

This says that nearby grades interact more strongly than grades that are far apart. The parameter $\sigma$ controls how broadly the coupling spreads across the hierarchy.

For small $\sigma$, coupling is sharply localized around neighboring grades. For large $\sigma$, more distant grades participate appreciably.

---

**Important scope statement**

The Gaussian is a clear and useful HSMT model choice. The present framework does not derive Gaussianity as the only possible law of intergrade coupling, and it does not derive a unique physical value of $\sigma$.

---

The reference value

$$\sigma_0 = \frac{\sqrt{2}}{4} \tag{13}$$

has been studied because it gives strong nearest-neighbor dominance in the corresponding discrete Gaussian model. It should be read as a reference-model parameter, not as an established constant of Nature.

## 6. What a single grade sees: exact projection

This is one of the most important parts of HSMT because it explains how a larger hierarchy can produce an effective theory inside one selected sector.

Let $P$ project onto the sector we want to describe and let

$$Q = I - P \tag{14}$$

project onto everything else.

The full Hamiltonian can then be viewed in block form,

$$H = \begin{pmatrix} PHP & PHQ \\ QHP & QHQ \end{pmatrix}. \tag{15}$$

The exact projected resolvent is the standard Feshbach–Schur expression [4, 5]

$$P(z - H)^{-1}P = \Big[z - PHP - PHQ(z - QHQ)^{-1}QHP\Big]^{-1}. \tag{16}$$

This defines an energy-dependent effective operator

$$H_{\text{eff}}(z) = PHP + \Sigma(z), \tag{17}$$

with self-energy

$$\Sigma(z) = PHQ(z - QHQ)^{-1}QHP. \tag{18}$$

The word *self-energy* here means the correction produced when the selected sector couples out to the hidden sector and then back again.

**A simple analogy**

Imagine studying one room of a building while the doors to other rooms remain open. You may only record what happens in your room, but sounds, airflow, and people moving elsewhere can still affect what you observe. Projection theory is the mathematical version of eliminating the unseen rooms while retaining their influence on the room you measure.

This effect can be exact. No claim of approximation is needed merely to write Eq. (16).

## 7. Memory and delayed influence

Projection has a time-domain interpretation as well. Once hidden degrees of freedom are removed, the visible sector need not evolve according to a simple instantaneous equation. Its present behavior can depend on its earlier state through a memory kernel, a structure familiar from projection-operator methods [6, 7, 8].

Schematically,

$$i\frac{\mathrm{d}}{\mathrm{d}t}|\psi_P(t)\rangle = PHP|\psi_P(t)\rangle + \int_0^t K(t-s)|\psi_P(s)\rangle\,\mathrm{d}s + \text{source term}. \tag{19}$$

The kernel $K$ records how amplitude that left the visible sector can influence it later.

This gives HSMT a natural language for effective nonlocality in time without making the microscopic theory itself nonunitary.

## 8. Dimension in HSMT

### 8.1. Grade is not dimension

HSMT deliberately keeps several ideas separate:

$$N \neq d, \qquad \text{shell} \neq \text{dimension}. \tag{20}$$

A grade tells us where a sector sits in the hierarchy. A dimension tells us how some geometric, spectral, or dynamical quantity scales.

Different notions of dimension may therefore coexist:

$$d_{\text{geom}}, \qquad d_{\text{res}}, \qquad d_{\text{spec}}, \qquad d_{\text{dyn}}. \tag{21}$$

Here the labels can be read as geometric, resolution-dependent, spectral, and dynamical dimension. Equality among them is not assumed.

### 8.2. Operational dimension from scaling

Suppose an observable quantity $Q(\ell)$ changes with a resolution or length scale $\ell$. A scale-dependent effective dimension can be defined by

$$d_Q(\ell) = \frac{\mathrm{d} \ln Q(\ell)}{\mathrm{d} \ln \ell}. \tag{22}$$

If $Q \propto \ell^d$ over some range, then Eq. (22) returns the exponent $d$ in that range.

This definition does not force a particular dimension into the theory. It measures a scaling behavior of a named quantity. Different observables can, in principle, produce different effective dimensions.

---

**Why this matters**

Instead of saying "this shell is four-dimensional" by definition, HSMT requires a dimension claim to be tied to a measurable or mathematically defined scaling law. This is a stricter standard and prevents the grade label from doing double duty as an unexplained dimension label.

---

## 9. Locality: what counts as nearby?

Geometry is not only about dimension. It is also about locality: which degrees of freedom count as neighbors and how strongly they interact.

A completely symmetric collection of degrees of freedom does not contain enough information to pick a unique neighborhood structure. If every element is interchangeable with every other element, the data themselves do not say why one pair should be called nearby and another pair distant.

HSMT therefore treats locality as an additional relational structure when it is needed.

One convenient representation is a graph or cell complex. Vertices represent basic sites or objects; edges represent declared local relations. The graph can carry data such as

$$(\ell_e, \mu_v, c_e), \tag{23}$$

where $\ell_e$ is an edge length, $\mu_v$ a measure associated with a vertex, and $c_e$ a conductance or coupling weight.

From this information one may build Laplacians, energies, diffusion processes, or Dirichlet forms in the standard mathematical sense [9, 10].

The key point is that locality is not generated merely by naming a grade. It must be encoded in relational data or derived from some future physical principle.

## 10. Metric, measure, and energy

To describe a geometric sector physically, one generally needs more than a list of neighboring sites.

A *metric* tells us about distance or separation. A *measure* tells us how size, volume, or weight is assigned. An *energy form* tells us how costly it is for a field or state to vary across the structure.

At an abstract level, HSMT can package these as

$$(d, \mu, \mathcal{E}). \tag{24}$$

This trio is useful because many continuum and discrete geometries can be discussed in the same language. It is also deliberately dimension-neutral: the framework does not have to decide the physical dimension before defining the basic geometric data.

For a graph-like realization, a quadratic energy may take the form

$$\mathcal{E}[f] = \frac{1}{2} \sum_{e=(u,v)} c_e \, |f(u) - f(v)|^2. \tag{25}$$

This says that rapid variation across strongly coupled neighboring sites costs more energy than slow variation.

## 11. Time and temporal evolution

The hierarchy label is not time. HSMT keeps time separate.

The basic temporal parameter is an affine variable $t$, and the full microscopic state evolves under Eq. (9). A particular physical model may later connect causal propagation, effective clocks, or arrows of time to additional structure, but those are not obtained simply by reading the grade number as a time coordinate.

This distinction matters because a sequence of grades and a sequence of moments answer different questions. Grade says *which sector*. Time says *when the state is evaluated*.

A reduced sector may display irreversible-looking behavior after hidden degrees of freedom are traced out or projected away, but microscopic unitary evolution alone does not guarantee a universal monotonic arrow of time. Diffusive or dissipative limits require additional dynamical conditions, such as suitable current decay or coarse-graining assumptions.

## 12. Refinement and changing descriptions

A geometric description may be refined: one coarse element can be replaced by several finer elements. Conversely, a fine description may be coarse-grained.

For HSMT, a refinement rule must do more than change a drawing. It has to say how states, measures, operators, and any physical inner products are transported between descriptions.

If $J$ maps a coarse state space into a refined state space, then a physically consistent refinement should preserve the relevant inner products after accounting for the measure used on each side. In a simple idealization,

$$\langle J\psi, J\phi\rangle_{\text{fine}} = \langle \psi, \phi\rangle_{\text{coarse}}. \tag{26}$$

Such rules are important if a shell is to be interpreted as a geometric regime rather than as a fixed finite graph.

## 13. Internal algebra and matter-like structure

Once locality and an inner product are given, one can construct standard algebraic operations on discrete forms or cochains. For example, exterior multiplication and its adjoint contraction can satisfy Clifford-like anticommutation relations when the appropriate positive inner product is supplied.

This can provide a mathematical seed for spinor-like or fermion-like structures, but a seed is not an identification.

**A necessary distinction**

Finding a Clifford algebra inside a model does not by itself derive the observed fermions of the Standard Model. Likewise, finding an internal symmetry does not by itself identify the physical gauge group. Additional physical structure and an operational bridge are required.

HSMT therefore treats matter constructions as conditional modules. They show that familiar algebraic structures can be supported inside the framework, but the observed particle content is not presently derived from the minimal HSMT architecture.

## 14. A controlled gravity-response branch

HSMT can also study how a geometric variable affects the energy of modes living on a relational structure.

Consider a positive operator

$$A(q) = m_f^2 I + B^T W(q) B, \tag{27}$$

where $q$ changes geometric coupling weights, $B$ is an incidence-type matrix, and $m_f$ provides a positive gap.

For the scaling

$$W(q) = e^q W_0, \tag{28}$$

the corresponding zero-point energy is

$$E_0(q) = \frac{1}{2} \operatorname{Tr} \sqrt{A(q)}. \tag{29}$$

For an individual positive edge contribution one obtains

$$\frac{\partial E_0}{\partial q_e} = \frac{1}{4} w_e \, b_e^T A^{-1/2} b_e > 0. \tag{30}$$

The result has a simple physical reading: in this tested positive branch, the vacuum energy changes monotonically with the scale variable instead of settling at a finite stationary point. The branch therefore supplies a controlled response calculation, but not a self-stabilizing vacuum.

This is useful information because it prevents a mathematically clean response kernel from being mistaken for a completed theory of gravity.

## 15. The optional P-CYCLE sector

P-CYCLE is an optional extension that places canonical degrees of freedom on independent cycles of a relational complex.

For a chain complex, the first homology space can be written

$$H_1 = Z_1 / B_1, \tag{31}$$

where $Z_1$ is the space of closed one-cycles and $B_1$ is the subspace of boundaries. Its dimension

$$b_1 = \dim H_1 \tag{32}$$

counts the number of independent cycle directions.

A quadratic cycle Hamiltonian is

$$H_{\text{cyc}} = \frac{\omega_c}{2} \left[ \pi^T G^{-1} \pi + h^T G h \right], \tag{33}$$

where $h$ is a cycle coordinate, $\pi$ its conjugate momentum, and $G$ a positive metric on cycle space.

After a canonical change of variables, the spectrum becomes

$$E_{\mathbf{n}}^{\text{cyc}} = \omega_c \left( |\mathbf{n}| + \frac{b_1}{2} \right). \tag{34}$$

The detailed metric $G$ does not alter these oscillator energy levels. Under the common metric scaling used in the tested vacuum branch, the cycle energies are scale-neutral.

Thus P-CYCLE is a coherent optional structural sector, but it does not supply the missing vacuum stabilization in that branch, and it is not presently identified with electromagnetism, a known gauge field, dark matter, or any other established physical field.

## 16. How HSMT could connect to experiment

A mathematical decomposition becomes physical only if experiments can identify it.

Suppose an experimental apparatus couples to the microscopic system through a contact map $B$. A natural response object is

$$M_B(z) = B^\dagger (z - H)^{-1} B. \tag{35}$$

This tells us how the full system responds when it is probed through the available contacts.

However, there is an important ambiguity. If a unitary transformation $U$ leaves all operational contacts and controls unchanged, then internal descriptions related by that transformation can be experimentally indistinguishable.

For several contacts $B_a$ and controls $C_\alpha$, the operationally invisible transformations form a stabilizer of the form

$$\mathcal{G}_{\mathrm{op}} = \left\{ U : UB_a = B_a,\ UC_\alpha U^\dagger = C_\alpha \right\}. \tag{36}$$

A proposed physical grading must therefore be well defined up to transformations that experiments cannot distinguish.

**Why this is the main physical challenge**

The mathematical projectors $P_\gamma$ are easy to write down once a grading is chosen. The hard question is why a particular family of projectors should represent real physical shells rather than one convenient way of organizing the Hilbert space.

## 17. What would make the grading physical?

A physical grading selector would be a rule that takes independently known physical structure and returns an equivalence class of grade projectors,

$$\mathscr{S} : \mathcal{X}_{\mathrm{phys}} \longrightarrow \mathrm{Grad}_\Gamma(\mathcal{H})/\mathcal{A}_\Gamma, \tag{37}$$

where $\mathcal{A}_\Gamma$ represents the allowed relabellings of the grade coordinate.

In ordinary language, such a rule must answer:

1. What physical data tell us where the grade boundaries are?

2. Why is that decomposition not merely a convenient basis choice?
3. Can the same rule be applied before looking at the result we want to predict?
4. Does the selected grading survive changes of representation that are invisible to the apparatus?
5. Does it leave some independent observable unused, so the theory can make an out-of-sample prediction?

A single projector is not enough. If one subspace is selected but the rest of the Hilbert space is left undivided, the complement can often be split into many inequivalent families of grades. A physical HSMT grading therefore needs a rule for the family, not merely a favored sector.

## 18. What would count as a distinct HSMT prediction?

Exact projection, self-adjoint dynamics, positive spectral measures, memory kernels, and graded Hilbert spaces are all useful, but they are not unique fingerprints of HSMT. Other quantum systems can exhibit them too.

For HSMT to become empirically distinctive, it needs two things together:

$$\boxed{\text{physically identifiable grading} + \text{non-generic measurable intergrade law}} \tag{38}$$

The second part matters because selecting a grading alone would still leave ordinary projection theory. HSMT must also impose some measurable relation among the selected sectors that a serious alternative theory need not satisfy.

The relation should be specified before the test data are inspected. Otherwise the same data could be used both to construct the theory and to declare it successful.

## 19. What HSMT presently says about familiar physics

This section is included to prevent ordinary words such as *dimension*, *matter*, and *gravity* from carrying more meaning than the equations support.

### 19.1. Four-dimensional spacetime

HSMT can represent grades and can define scale-dependent dimensions, but the present framework does not derive that one particular grade must have four spacetime dimensions. A physical derivation would need a selector and an independently defined dimensional observable.

### 19.2. The Standard Model

The framework can support internal algebras, Clifford-like structures, and other mathematical ingredients useful in particle physics. It does not presently derive the observed Standard Model gauge group, three generations, Yukawa couplings, Higgs parameters, or neutrino data.

### 19.3. Gravity

The framework contains controlled geometric-response calculations. The tested positive vacuum branch does not stabilize itself, and no derivation of Einstein's equations from the core is presently established.

### 19.4. Cosmology

Ideas involving redshift, dark matter, dark energy, the cosmic microwave background, or transitions between large-scale shells require an operational bridge that has not yet been supplied. They are not current predictions of the theory.

## 20. What HSMT does provide now

The absence of a completed empirical identification does not make the framework empty. HSMT provides a definite mathematical architecture with several controlled consequences.

It provides:

- a precise graded state space;
- a clean separation between grade and dimension;
- ordinary self-adjoint microscopic dynamics;
- explicit intergrade coupling models;
- exact Feshbach projection to an effective sector;
- exact self-energy and memory structures generated by hidden sectors;
- a framework for scale-dependent dimensional observables;
- explicit relational locality and metric-measure-energy modules;
- a consistent separation of grade and time;
- conditional internal-algebra and matter-like constructions;
- controlled geometric-response calculations;
- an optional cycle sector with an exact oscillator spectrum;
- an operational test for whether a proposed grading is physically identifiable.

These ingredients make HSMT a framework in which questions about hidden geometric sectors can be posed without confusing model assumptions with established physical facts.

## 21. A worked conceptual example

Consider a simplified hierarchy with three grades,

$$\mathcal{H} = \mathcal{H}_{-1} \oplus \mathcal{H}_0 \oplus \mathcal{H}_{+1}. \tag{39}$$

Suppose an experiment can directly probe only the middle sector $\mathcal{H}_0$. The full Hamiltonian may be written schematically as

$$H = \begin{pmatrix} H_{-1} & V_{-1,0} & 0 \\ V_{0,-1} & H_0 & V_{0,+1} \\ 0 & V_{+1,0} & H_{+1} \end{pmatrix}. \tag{40}$$

The observer in the middle grade does not simply see $H_0$. Exact elimination of the two neighboring grades produces

$$H_{\text{eff}}(z) = H_0 + V_{0,-1}(z - H_{-1})^{-1}V_{-1,0} + V_{0,+1}(z - H_{+1})^{-1}V_{+1,0}. \tag{41}$$

This small model captures the central HSMT mechanism. Hidden grades can shift energy levels, create resonant structure, change response functions, and generate memory when the description is reduced to the visible grade.

What it does *not* tell us is whether $\mathcal{H}_0$ corresponds to our physical spacetime, whether the neighboring sectors are geometric dimensions, or whether the couplings have a Gaussian form. Those questions require additional physical input.

## 22. How to read future HSMT models

A clear HSMT model should identify each of the following explicitly:

1. **The grades:** What is the set $\Gamma$, and what does one grade represent?
2. **The state spaces:** What is $\mathcal{H}_\gamma$ on each grade?
3. **The geometry:** What metric, measure, graph, manifold, or energy structure is assigned?
4. **The microscopic operator:** What is $H$, and on what domain is it self-adjoint?
5. **The coupling law:** Which terms connect different grades?
6. **The visible sector:** What projector $P$ represents the experimentally accessible part?
7. **The effective theory:** What follows from exact or controlled reduction?
8. **The operational bridge:** How would an experiment identify the proposed grading?
9. **The distinctive law:** What measurable relation is not generic quantum mechanics?
10. **The comparator:** What credible alternative could fail where the HSMT model succeeds?

This checklist keeps technical richness from becoming conceptual ambiguity.

## 23. Current physical boundary

The present framework reaches a clear boundary.

**Current boundary of the theory**

HSMT has a coherent graded-geometric architecture and exact sector-reduction machinery, but it does not yet contain a physical principle that uniquely identifies the grading realized in Nature together with a uniquely HSMT-specific measurable relation among the selected grades.

This boundary is constructive. It tells us what kind of new result would materially advance the theory.

A future completion would need an independently motivated selector, an operational bridge, and a prediction that survives comparison with conventional models. Adding more mathematical sectors without solving that identification problem would increase the size of the framework without increasing its physical predictivity.

## 24. Conclusion

Hierarchical Shell Manifold Theory describes physics through a hierarchy of geometric sectors whose quantum state spaces can be coupled. The hierarchy is represented by a graded Hilbert space, the full dynamics are generated by a self-adjoint operator, and exact projection determines what a selected sector experiences when the remaining sectors are not directly observed.

The theory's central physical possibility is that effective behavior inside one sector may carry traces of coupling to other sectors. Those traces can appear as energy-dependent self-energies, memory terms, modified spectra, or scale-dependent observables. HSMT also provides modules for locality, metric and measure, internal algebra, geometric response, and optional cycle degrees of freedom.

The language is intentionally careful. A shell is a hierarchical regime, not necessarily a literal spatial shell. A grade is a label, not automatically a dimension. An effective dimension is a scaling measurement, not a number assigned by definition. A mathematical symmetry is not automatically a Standard Model gauge symmetry. A geometric response is not automatically Einstein gravity. An engineered analog is not automatically evidence that Nature uses the same ontology.

The remaining physical challenge can therefore be stated simply: identify the real grading, and then find a measurable relation between its sectors that ordinary quantum mechanics does not already guarantee. If that step is achieved, HSMT would move from a controlled framework for graded effective physics toward a distinctive physical theory.

## A. Plain-language glossary

| Term | Meaning in this paper |
|---|---|
| Grade | A label identifying one sector in the hierarchy. It is not automatically a physical dimension. |
| Shell | A coherent hierarchical regime associated with a grade. It need not be a literal spherical shell. |
| Manifold | A geometric space that can be described locally by coordinates. |
| Hilbert space | The vector space used to describe quantum states and their inner products. |
| Projector | An operator that extracts the component of a state belonging to a chosen sector. |
| Self-adjoint operator | An operator with the mathematical properties required for standard unitary quantum evolution. |
| Intergrade coupling | A term in the dynamics that connects different grades. |
| Effective theory | The description seen after degrees of freedom outside a selected sector are eliminated or ignored in a controlled way. |
| Resolvent | The operator $(z - H)^{-1}$, useful for studying spectra and response. |
| Self-energy | The correction to a selected sector produced by leaving that sector, propagating through hidden degrees of freedom, and returning. |
| Memory kernel | A time-domain term through which the present reduced dynamics depend on earlier states. |
| Spectral dimension | A dimension inferred from the scaling of a spectral or diffusion quantity. |
| Metric | A rule for assigning distance or separation. |
| Measure | A rule for assigning size, volume, or weight. |
| Dirichlet form | A quadratic energy that measures how rapidly a function varies across a space or graph. |
| Homology cycle | A closed topological path that is not merely the boundary of a higher-dimensional piece. |
| Operational bridge | A rule connecting internal mathematical objects to actual preparations, controls, and measurements. |
| Stabilizer | A set of internal transformations that leave the available experimental operations unchanged. |
| Selector | A physical rule that identifies which mathematical grading corresponds to the grading realized by the system. |

## B. Compact mathematical summary

The core graded state space is

$$\mathcal{H} = \bigoplus_{\gamma \in \Gamma} \mathcal{H}_\gamma, \qquad \sum_\gamma P_\gamma = I. \tag{42}$$

Microscopic evolution is

$$H = H^\dagger, \qquad U(t) = e^{-itH}. \tag{43}$$

For a selected sector $P$ and complement $Q = I - P$,

$$H_{\text{eff}}(z) = PHP + PHQ(z - QHQ)^{-1}QHP. \tag{44}$$

A reference Gaussian coupling is

$$K_{NM} = K_0 \exp\left[-\frac{(N-M)^2}{2\sigma^2}\right]. \tag{45}$$

A scale-dependent dimensional observable is

$$d_Q(\ell) = \frac{\mathrm{d}\ln Q(\ell)}{\mathrm{d}\ln \ell}. \tag{46}$$

A representative metric-measure-energy package is

$$(d, \mu, \mathcal{E}). \tag{47}$$

The optional P-CYCLE spectrum is

$$E^{\text{cyc}}_{\mathbf{n}} = \omega_c \left(|\mathbf{n}| + \frac{b_1}{2}\right). \tag{48}$$

The missing empirical step is

$$\boxed{\text{physical grading selector} + \text{non-generic measurable intergrade law.}} \tag{49}$$

## References


[1] V. M. Garrett, *Hierarchical Shell-Manifold Theory: A Matured Foundational and Reference-Model Framework*, HSMT manuscript, September 2026.

[2] V. M. Garrett, *Hierarchical Shell Manifold Theory: Operator Reconstruction, Spectral Identifiability, and No-Go Boundaries*, HSMT manuscript, 2026.

[3] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer-Verlag, Berlin, 1976; reprinted in Classics in Mathematics, Springer, 1995.

[4] H. Feshbach, "Unified theory of nuclear reactions," *Annals of Physics* **5**, 357–390 (1958), doi:10.1016/0003-4916(58)90007-1.

[5] H. Feshbach, "A unified theory of nuclear reactions. II," *Annals of Physics* **19**, 287–313 (1962), doi:10.1016/0003-4916(62)90221-X.

[6] S. Nakajima, "On Quantum Theory of Transport Phenomena: Steady Diffusion," *Progress of Theoretical Physics* **20**, 948–959 (1958), doi:10.1143/PTP.20.948.

[7] R. Zwanzig, "Ensemble Method in the Theory of Irreversibility," *Journal of Chemical Physics* **33**, 1338–1341 (1960), doi:10.1063/1.1731409.

[8] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press, Oxford, 2002.

[9] F. R. K. Chung, *Spectral Graph Theory*, CBMS Regional Conference Series in Mathematics, Vol. 92, American Mathematical Society, Providence, 1997.

[10] M. Fukushima, Y. Oshima, and M. Takeda, *Dirichlet Forms and Symmetric Markov Processes*, De Gruyter Studies in Mathematics, Vol. 19, Walter de Gruyter, Berlin, 1994.